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UK/US Market Access Divergence in HER2-targeted breast cancer therapies

Section 1: Abstract

This report uses trastuzumab deruxtecan (T-DXd) as a case study in UK/US market access divergence, drawing on a verified n=13 NICE appraisal dataset, a four-document theme comparison of NICE and US cost-effectiveness literature, and a single-medicine, two-indication case study of T-DXd itself. Four findings are reported. EAG disagreement is the strongest observed predictor of NICE outcome in the verified dataset, though at n=13 this should be interpreted as hypothesis-generating rather than validated. Surrogate endpoint validity is a live point of contest in NICE appraisals and a categorical absence in US academic CEAs, a structural difference rather than a matter of emphasis. A positive reimbursement outcome for one indication carries no guarantee for another, even where medicine, manufacturer, and appraising body are held constant. NICE's most generous historical threshold, converted to USD, remains below the floor of the US benchmark range, a gap not attributable to currency or reporting convention. Each finding carries its own stated limitation and should not be generalised beyond the evidence base that produced it. Where a predictive model was attempted and found unstable at the available sample size, that instability is reported as an independent finding than smoothed over. Combined, the findings suggest UK market access risk is structural rather than incidental, and cheapest to address at the evidence-generation stage of a launch rather than after a US-anchored strategy is already set.

Section 2: Case Background

2.1: Disease and treatment context

HER2 (human epidermal growth factor receptor 2) is a cell-surface receptor that, when over-expressed, drives an aggressive subtype of breast cancer historically associated with poor prognosis relative to other subtypes. HER2-positive disease is estimated to affect roughly one in five breast cancer patients[1].

Since the introduction of trastuzumab in the late 1990s, HER2-targeted therapy has substantially improved outcomes for this group, with successive agents, including

[1] OncLive (2021) NICE Recommends Trastuzumab Deruxtecan in Advanced HER2+ Breast Cancer. Available at: https://www.onclive.com/view/nice-recommends-trastuzumab-deruxtecan-in-advanced-her2-breast-cancer

pertuzumab and trastuzumab emtansine (T-DM1), extending the treatment pathway for patients whose disease progresses on earlier lines[2].

A separate and more recently defined population, HER2-low disease, describes tumours with some HER2 expression (IHC 1+, or IHC 2+ with a negative ISH result) that falls short of the threshold for HER2-positive classification[3]. Under prior treatment approaches, HER2-low tumours were treated as HER2-negative and were not eligible for HER2-targeted therapy. This case study is built around a single medicine that was subsequently approved for both populations, which is what allows Sections 3.3 and 3.4 to compare access outcomes for the same drug across two related but distinct indications.

2.2 The medicine: trastuzumab deruxtecan

Trastuzumab deruxtecan (T-DXd, brand name Enhertu) is an antibody-drug conjugate (ADC) combining a HER2-targeting monoclonal antibody with a topoisomerase I inhibitor payload, joined by a cleavable linker[4]. After the antibody binds HER2 on the tumour cell surface, the conjugate is internalised and the payload is released intra-cellularly, inducing DNA damage and apoptosis[5]. T-DXd carries a comparatively high drug-to-antibody ratio and exhibits a "bystander effect," in which the released payload can also act on neighbouring tumour cells with lower HER2 expression, a mechanism proposed as part of the rationale for its efficacy in HER2-low disease specifically [6]. T-DXd is jointly developed and commercialised by Daiichi Sankyo and AstraZeneca.

2.3 US regulatory pathway

T-DXd's US regulatory history proceeded through a sequence of expanding indications, three of which are directly relevant to this analysis:

- 1) December 2019: accelerated approval for HER2-positive unresectable or metastatic breast cancer after two or more prior anti-HER2 regimens, based on the single-arm DESTINY-Breast01 trial[7].

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[2] Saleh, K., Khoury, R., Khalife, N., Chahine, C., Ibrahim, R., Tikriti, Z. and Le Cesne, A. (2024) 'Mechanisms of action and resistance to anti-HER2 antibody-drug conjugates in breast cancer', *Cancer Drug Resistance*. doi: 10.20517/cdr.2024.06.

[3] Targeted Oncology (2026) FDA Approves T-DXd in HR+, HER2-Low/Ultralow Breast Cancer.

[4] ScienceDirect (n.d.) Trastuzumab Deruxtecan – an overview.

[5] ClinicalTrials.gov (NCT04014075) DS-8201a in HER2-positive Gastric Cancer: protocol and statistical analysis plan.

[6] ClinicalTrials.gov (NCT03334617) Phase II Umbrella Study of Novel Anti-cancer Agents in Participants With NSCLC Who Progressed on an Anti-PD-1/PD-L1 Containing Therapy: protocol. Available at: https://cdn.clinicaltrials.gov/large-docs/17/NCT03334617/Prot_000.pdf

[7] Narayan, P. et al. (2021) 'FDA Approval Summary: Fam-trastuzumab deruxtecan-nxki for the treatment of unresectable or metastatic HER2-positive breast cancer', *Clinical Cancer Research*. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC8570919/

- 2) May 2022: conversion to regular (full) approval for HER2-positive disease after one or more prior anti-HER2 regimens, based on the confirmatory randomised DESTINY-Breast03 trial, which compared T-DXd against T-DM1 in the second-line setting[8]. This is the approval referenced as the US milestone in Section 3.3's timeline analysis.
- 3) August 2022: approval for HER2-low metastatic breast cancer following prior chemotherapy, based on the randomised DESTINY-Breast04 trial, marking the first approval of a HER2-targeted therapy for a population previously classified as HER2-negative[9].

Later approvals, including a January 2025 extension to HER2-low and HER2-ultralow disease after endocrine therapy (DESTINY-Breast06) and 2025–2026 approvals in earlier-line and early-stage HER2-positive settings, fall outside the appraisal window examined in this analysis and are noted here only for completeness[10].

2.4 UK regulatory pathway and the two appraisals

NICE evaluates new medicines through its Single Technology Appraisal (STA) process, in which the manufacturer submits a cost-effectiveness case that is independently evaluated by an Evidence Review Group (ERG) before an appraisal committee reaches a final recommendation[11]. Two STAs for T-DXd form the basis of this analysis:

- **1) TA862 (HER2-positive, second-line), Published 1 February 2023:** TA862 recommended T-DXd with managed access (via the Cancer Drugs Fund) for HER2-positive unresectable or metastatic breast cancer after one or more prior anti-HER2 treatments, drawing on DESTINY-Breast03 data[12]. The committee's stated rationale centred on immature overall survival data at the time of appraisal. Progression-free survival benefit was accepted, but because the confirmatory trial was still ongoing, the committee judged that cost-effectiveness estimates carried too much uncertainty for a routine commissioning recommendation, and instead deferred a final decision pending further survival data collected under managed access[13].

[8] FDA (2022) FDA grants regular approval to fam-trastuzumab deruxtecan-nxki for breast cancer. 4 May. Available at: https://www.fda.gov/drugs/resources-information-approved-drugs/fda-grants-regular-approval-fam-trastuzumab-deruxtecan-nxki-breast-cancer
[9] CancerNetwork (2025) T-DXd Approval Appears Significant in HER2-Low/Ultralow Breast Cancer.
[10] AstraZeneca (2025) Enhertu approved in the US as first HER2-directed therapy for patients with HER2-low or HER2-ultralow metastatic breast cancer.
[11] Liverpool Reviews and Implementation Group (2023) Trastuzumab Deruxtecan for Treating HER2-Positive Unresectable or Metastatic Breast Cancer After Two or More Anti-HER2 Therapies: An ERG Perspective of a NICE STA.
[12] NICE (2023a) Trastuzumab deruxtecan for treating HER2-positive unresectable or metastatic breast cancer after 1 or more anti-HER2 treatments (TA862).
[13] NICE (2023b) TA862: 1 Recommendations.

- **TA992 (HER2-low):** This appraisal considered T-DXd for HER2-low metastatic or unresectable breast cancer, drawing on DESTINY-Breast04 data[14]. The committee did not recommend T-DXd within its marketing authorisation for this indication, and no managed access route was agreed.

The two appraisals share the same medicine, the same manufacturer, and the same appraising body, differing only in the indication under review and, therefore, the trial evidence base and committee outcome. This is the pairing that Section 3.3 uses to isolate indication as the operative variable in reimbursement outcome, and that Section 3.4 uses alongside the US ICER benchmark to frame the wider UK/US cost-effectiveness threshold gap.

2.5: Case selection reasoning

T-DXd is a useful case study for a market access comparison precisely because it removes several cofounding factors that would otherwise complicate a UK/US comparison. Manufacturer strategy, drug pricing methodology, and underlying clinical mechanism are held constant across both jurisdictions and across both indications examined. What varies is the regulatory and reimbursement environment the same evidence package is submitted into. However, this is also the source of the case's central limitation, carried forward from Section 3.3. Specifically, findings drawn from one medicine's two indications illustrate frictions that may recur elsewhere in NICE's broader appraisal activity, but do not by themselves establish how often they do.

This analysis is scoped to the UK and US systems specifically. The institutional mechanisms identified here (NICE's single-payer threshold logic and the fragmented US landscape of ERISA-preempted self-funded plans, CMS reimbursement, and IRA negotiation authority), do not generalise to other jurisdictions without further comparative work. For instance, Australia's PBS pairs a cost-effectiveness threshold (similar to that of NICE) with managed entry agreements that, in practice, rely mainly on confidential price discounts rather than outcomes data[15], proposing a middle ground on the evidence-versus-price spectrum rather than a simple UK/US binary.

2.6 Institutional context for UK/US divergence

The threshold gap quantified in Section 3.4 reflects more than a difference in stated willingness-to-pay. It reflects two structurally different payer systems. NICE operates as a

[14] NICE, Technology Appraisal Guidance TA992, "Trastuzumab deruxtecan for treating HER2-low metastatic or unresectable breast cancer after chemotherapy," published 29 July 2024.
[15] Robinson, M.F., Mihalopoulos, C., Merlin, T., & Roughead, E. (2018). Characteristics of managed entry agreements in Australia. *International Journal of Technology Assessment in Health Care*, 34(1), 46–55. https://doi.org/10.1017/S0266462317001106

single, budget-constrained evaluator inside a taxpayer-funded NHS, ultimately accountable to Parliament and the Treasury for aggregate health spend,[16] which is why a positive NICE recommendation is legally binding and must be funded, while a negative recommendation means the treatment will not be routinely commissioned, and a budget impact breach triggers mandated commercial negotiations rather than automatic rejection.[17] The US market, by contrast, has no equivalent single payer or mandated cost-effectiveness gate.[18] Reimbursement decisions are dispersed across private insurers, Medicare, and Medicaid,[19] none of which is bound by ICER's benchmark, and ICER's range functions as a widely cited reference point rather than an enforceable rule,[20] precisely because no US body holds the position NICE holds in the UK.[21] This distinction is not incidental to the numbers in Section 3.4. It is the reason those numbers behave differently. A UK threshold breach and a US benchmark breach are not two instances of the same event measured in different currencies, but two different kinds of event, one with a binding institutional consequence and one without.

The US system's fragmentation is not merely a policy default, rather, it is a product of specific legal architecture. ERISA preempts state regulation of self-funded employer health plans, which is why large employer coverage (a substantial share of the commercially insured population) sits structurally outside any single state or federal cost-effectiveness restriction.[22] Medicare itself faces a parallel statutory constraint. Not only is CMS barred from using QALY-based evidence in ordinary coverage and reimbursement determinations (see n.18, above), but when the IRA created Medicare's Drug Price Negotiation Program in 2022, Congress wrote the identical prohibition directly into the negotiation statute itself, the Secretary "shall not use evidence from comparative clinical effectiveness research in a manner that treats extending the life of an elderly, disabled, or terminally ill individual as of lower value than extending the life of an individual who is younger, nondisabled, or not terminally ill."[23] This means the IRA's move toward centralized price setting, which initially appears to be the US system's nearest equivalent to NICE, was deliberately built without the

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[16] DHSC & NICE, Framework Agreement 2018, paras. 4.1–4.2.

[17] NHS England, NHS Commercial Framework for New Medicines (2025), paras. 78, 81, 134.

Mulligan et al., *Health Technology Assessment for the U.S. Healthcare System*, USC Schaeffer Center (2020), p. 1.

Mulligan et al. (2020), p. 1, on the US's decentralized insurance system. Medicare specifically is under a statutory bar on using QALY-based comparative-effectiveness evidence in coverage or reimbursement determinations (Social Security Act §1182(c)(1), 42 U.S.C. §1320e-1(c)(1), added by the ACA). This is a stronger constraint than simple non-adoption of ICER's benchmark.

Mulligan et al. (2020); Enright et al., *JMCP* 31(3) (2025), pp. 291–292. Across 17 US commercial health plans' publicly available specialty drug coverage policies, HTAs accounted for only 3.2% of all cited clinical/economic evidence (450 of 14,033 citations); among HTA citations specifically, ICER was the leading US-based source (75 citations, 17.7%), behind NICE (130, 30.7%) as the single most-cited organisation overall.

Mulligan et al. (2020), p. 1; Enright et al. (2025), Discussion

ERISA, 29 U.S.C. §1144(a) (preemption of state regulation of self-funded employee benefit plans).

[23] Social Security Act §1194(e)(2), 42 U.S.C. §1320f-3(e)(2), added by the Inflation Reduction Act, Pub. L. 117-169, title I, §11001(a) (Aug. 16, 2022), 136 Stat. 1843, 1854.

cost-effectiveness threshold logic that defines NICE's role. Where NICE's mandate and its QALY-based methodology are the same instrument, the Drug Price Negotiation Program's authority to set prices and its statutory prohibition on QALY-based reasoning are two halves of the same design choice. The result is not an absence of centralisation in US drug pricing but centralisation built specifically to exclude the mechanism that gives NICE's threshold its institutional force.

Section 3: Analysis

3.1 Diagnostics: Predictors of NICE Appraisal Outcome

Headline finding

Across the verified n=13 population of HER2-positive and HER2-low breast cancer therapy appraisals, the strongest observed predictor of NICE outcome is disagreement between the company's model and the Evidence Assessment Group (EAG), coded as an ordinal variable ("eag_disagreement_ord", aligned = 0, partial disagreement = 1, substantial disagreement = 3)

EAG disagreement	n	Outcome pattern
0 (aligned)	5	100% recommended
1 (partial)	3	2 recommended, 1 CDF managed access
2 (substantial)	5	4 not recommended, 1 CDF managed access

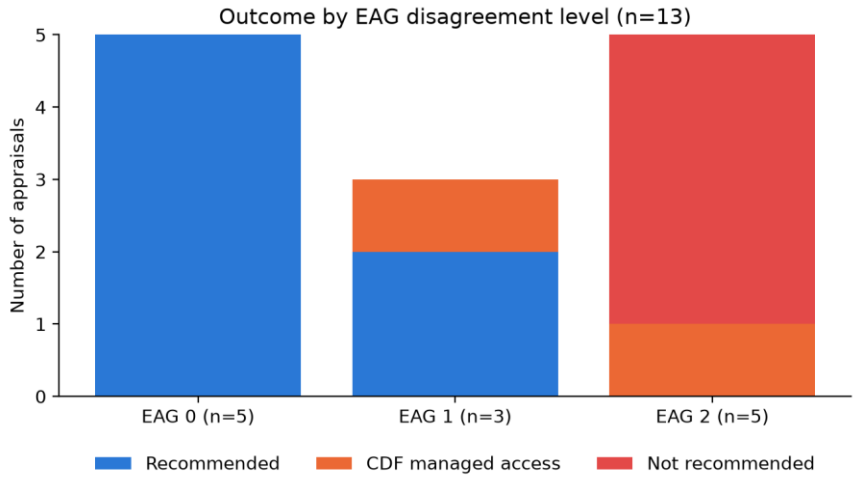


Figure 1: 3.1_eag_separation_chart.png (outcome by EAG disagreement level, n labelled per group.)

No appraisal with substantial EAG disagreement received a routine recommendation, and no appraisal with an aligned EAG assessment was rejected or referred to managed access. The pattern is monotonic and directionally consistent with the underlying mechanism, whereby a greater analytical disagreement between company and assessor is followed by a worse committee outcome.

Limitation

Given a total of only 13 observations, drawn from cells comprising of 3–5 observations each, this pattern is best interpreted as preliminary evidence. It suggests that EAG disagreement is meaningfully associated with NICE decision-making outcomes and should be investigated further with a larger sample but should not be interpreted as an empirically validated predictive relationship.

Why a regression-only approach was not adopted

A logistic model fit on the full available covariate set (n=11, after excluding the two CDF-managed-access rows from the binary outcome) exhibits complete separation with 100% in-sample accuracy, coefficients diverging to roughly −16 for “eag_disagreement_ord” and +10 for “end_of_life” (intercept +13.6). This near-perfect in-sample fit is itself a diagnostic of overfitting rather than a sign the model performs well with only 11 observations and 2 features, the model has enough flexibility to draw a boundary that separates the two outcome classes exactly, and no held-out test set exists at this sample size to check whether that boundary generalises.

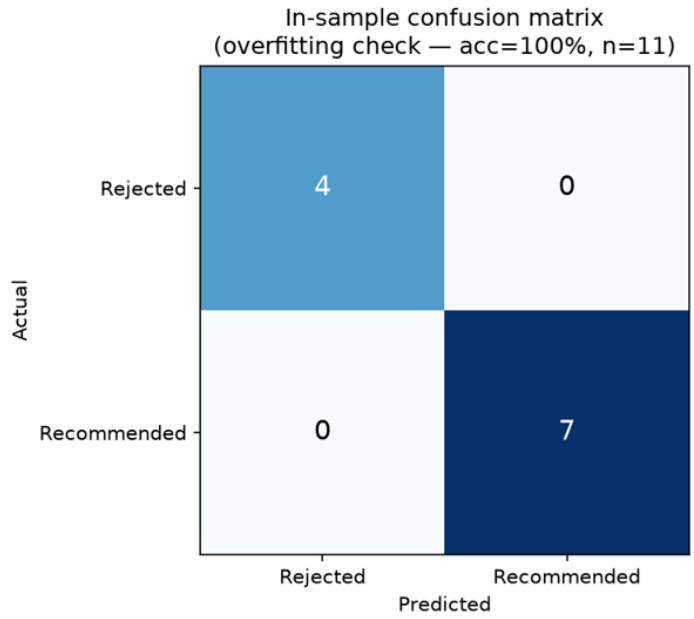


Figure 2a: confusion_matrix.png (in-sample classification results for the n=11 model.)

This reflects genuine instability rather than a robust effect, demonstrated by leave-one-out (LOO) sensitivity analysis. Across all 11 single-observation refits, coefficient magnitude exhibits considerable variation, with “eag_disagreement_ord” ranging from −8.4 to −16.3 and end_of_life from +7.8 to +11.2, while sign remains constant throughout. The direction of each effect is therefore stable, but its magnitude is highly sensitive to the exclusion of any single case, with swings of 4–8 units on coefficients whose baseline values lie in the 10–16 range.

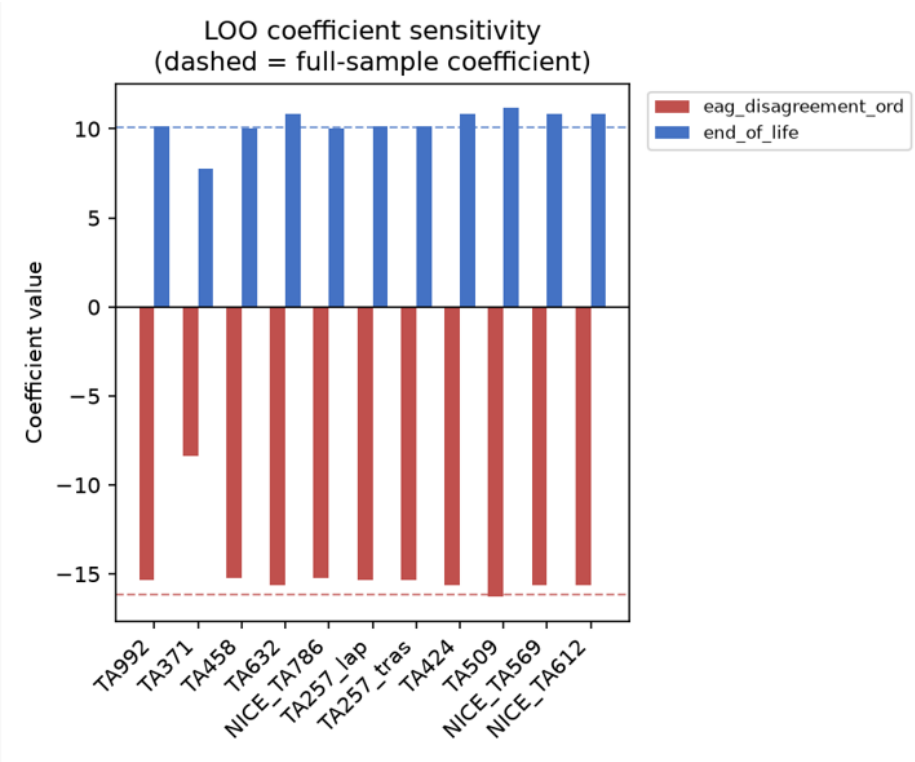


Figure 2b: loo_sensitivity.png (leave-one-out coefficient sensitivity across all 11 refits.)

The scale of the shortfall can be stated directly. Standard events per variable (EPV) guidance calls for 10–20 minority class events per predictor. Against the 4 rejected appraisals currently available in this sample, a loose exploratory target (5 EPV, 10 rejected appraisals needed) implies a shortfall of 6 cases, whereas a conservative target (10 EPV, 20 rejected appraisals needed) implies a shortfall of 16. Assuming NICE's typical 10–15% base rejection rate (the working assumption used in this calculation), reaching the loose target alone would require a total dataset on the order of 67–100 appraisals. The conservative target would require roughly 134–200, up to fifteen times the current n=13.

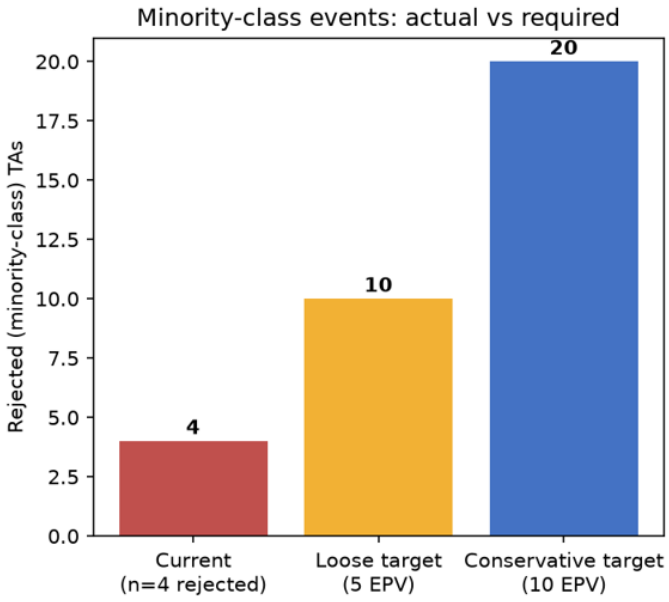


Figure 2c: epv_summary.png (events-per-variable shortfall and implied minimum dataset size.)

This is the expected behaviour of a maximum-likelihood model at this sample size when a predictor ("eag_disagreement_ord" in this case) separates the outcome cleanly, and it is reported as a methodological finding in its own right. It highlights the small-sample structure of NICE technology appraisal data and explains why the appropriate use of this dataset is descriptive and hypothesis generating rather than inferential, because a model this sensitive to individual observations cannot support a claim about effect size, even where the qualitative direction of the relationship is well-behaved.

Variables Not Supported as Independent Predictors

Two variables were tested and deemed unreliable as independent predictors:

- **"uplift_applied"** (whether an end of life or severity weighting was applied) shows perfect separation against outcome. This is circular by construction; the variable is coded from the outcome itself (an "insufficient" uplift is only labelled as such when the outcome was still negative). It is retained in the dataset for transparency but excluded from any claim about predictive relationships.
- **"end_of_life"** alone, independent of EAG disagreement, shows no meaningful separation against outcome (57% recommended where absent versus 75% where present, on n=11. This is a modest skew, not a signal). Combined with the finding above, this suggests that EAG disagreement, rather than EoL/severity status necessarily, is the primary driver of the observed outcome pattern in this sample.
- **"era"** (pre- vs post-2022 EoL methodology) has too small a post-2022 cell to support any claim and is noted, not interpreted.

Limitation (data completeness)

Only 5 of 13 appraisals carry a usable single point base case ICER; the remainder are confidential, not disclosed, reported only as a lower bound, or genuinely multi valued (see Section 3.1 data appendix). This is a feature of NICE's public disclosure practice, not a gap in extraction. Over half of published appraisals in this sample do not release a usable point estimate, and this is noted here because it constrains any future analysis that would use ICER magnitude as a covariate.

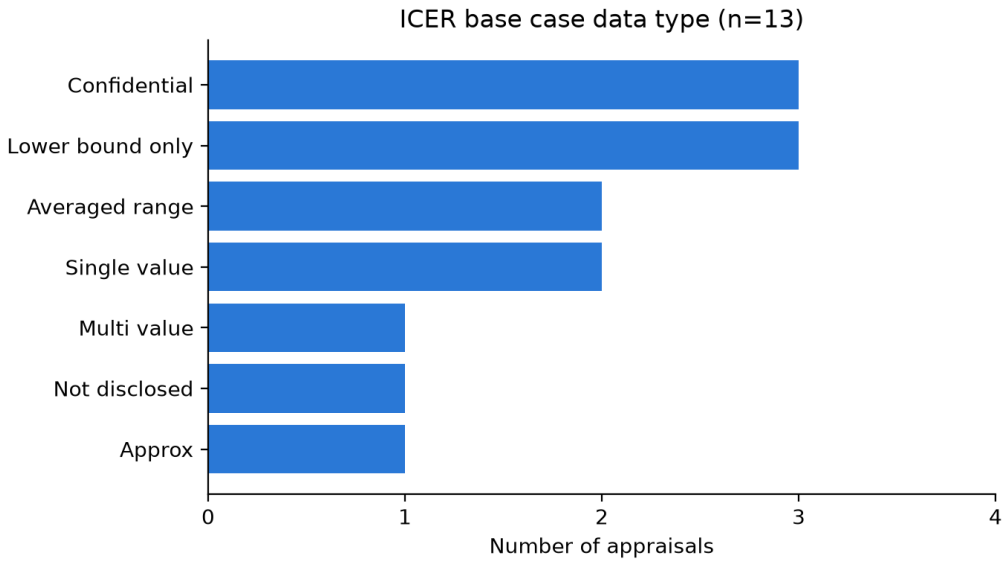


Figure 3: 3.1_icer_completeness_chart.png (count of appraisals by ICER base-case data type.)

Summary of findings

At n=13, this dataset supports one credible and transparent finding, specifically that EAG disagreement is associated with appraisal outcome, and two clearly documented negative results. The value of this diagnostic exercise resides as much in clarifying the inferential limits of a small, carefully verified dataset as in the substantive finding itself.

3.2 Payer Resistance Themes: NICE vs. US Academic Cost-Effectiveness Literature

Method

Eight resistance and uncertainty themes (comparator selection, survival data immaturity, population generalizability, cost effectiveness threshold ambiguity, structural and model assumptions, subgroup uncertainty, surrogate endpoint validity, and utility and QoL

uncertainty) were extracted via keyword matching from four source documents. These were two NICE technology appraisals for Enhertu (HER2 low, not recommended, 2024; HER2 positive, managed access, 2022) and two US academic cost effectiveness analysis papers (Peng et al.[24], Clinical Therapeutics; Shi et al.[25], PLOS ONE). Frequencies are normalized per 1,000 words to control for document length, with UK documents at 3,955 and 3,787 words and US papers at 1,067 and 1,015 words.

Headline finding: US models structurally bypass surrogate endpoint contestation

Surrogate endpoint validity (PFS to OS extrapolation) appears in both NICE appraisals (0.25 and 1.32 occurrences per 1,000 words) but registers zero mentions in either US paper. This is not a difference in emphasis, but a categorical absence. NICE's HTA process requires appraisal committees to assess whether a trial's primary endpoint (progression free survival) reliably predicts the outcome that matters for reimbursement (overall survival). US academic CEAs, built on published trial data and standard modelling frameworks, do not appear to treat this as a distinct point of contest at all.

[24] Peng, Y., Wang, L., Peng, L., Liu, Q., Yi, L., Luo, X., Li, S., Qin, S., Wan, X., Tan, C. and Chen, G. (2023) 'Cost-effectiveness of Trastuzumab Deruxtecan for HER2-low Advanced Breast Cancer in the United States', *Clinical Therapeutics*, 45(10), pp. 965–972. doi: 10.1016/j.clinthera.2023.07.014.
[25] Shi, D., Liang, X., Li, Y. and Chen, L. (2023) 'Cost-effectiveness of trastuzumab deruxtecan for previously treated HER2-low advanced breast cancer', *PLOS ONE*, 18(8), e0290507. doi: 10.1371/journal.pone.0290507.

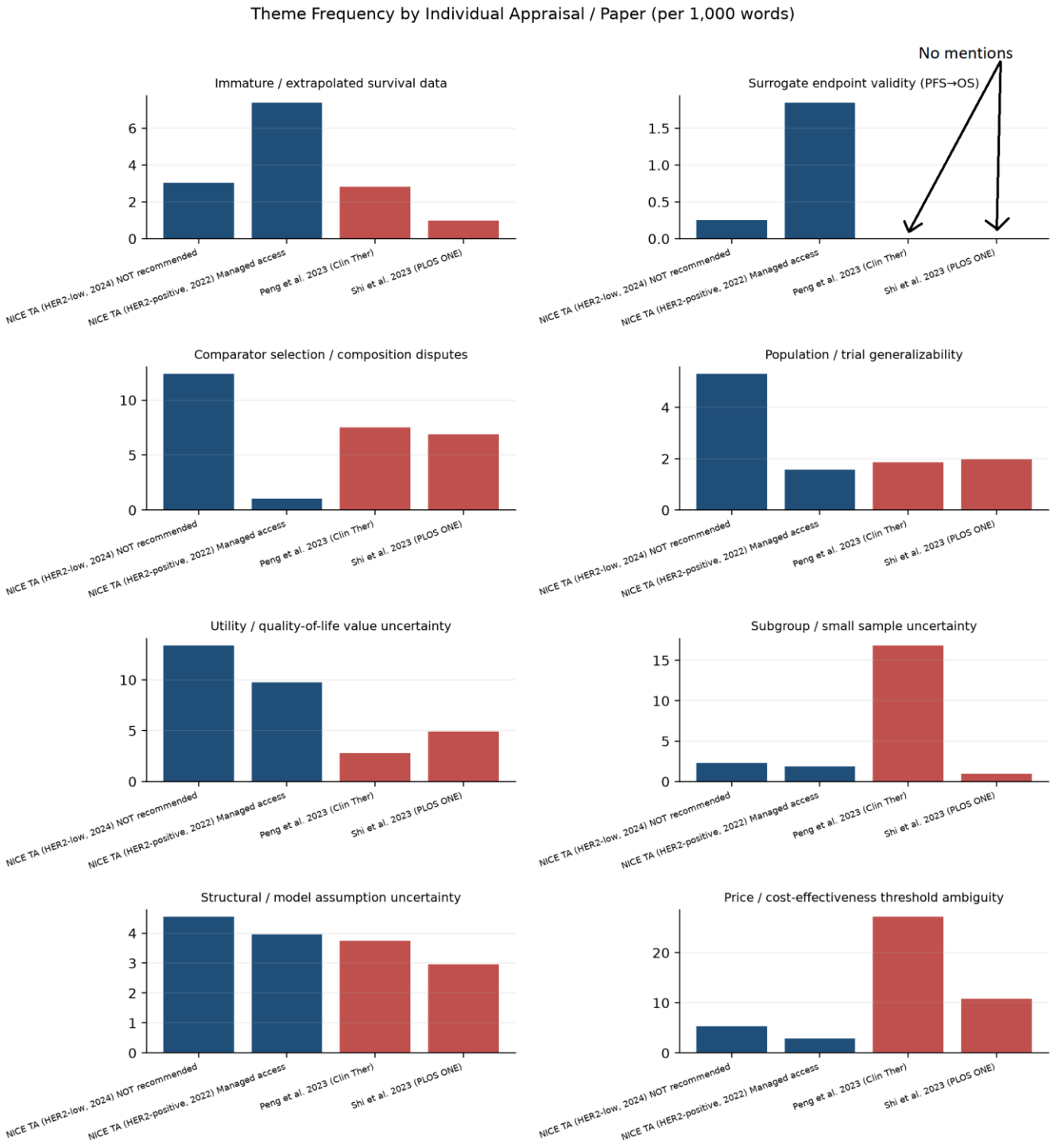


Figure 4: 3.2_chart_small_multiples_by_document.png. Theme frequency by document. Surrogate endpoint validity (annotated) is the only theme with zero mentions in both US papers (a categorical absence, not a magnitude difference)

Aggregated UK vs. US pattern

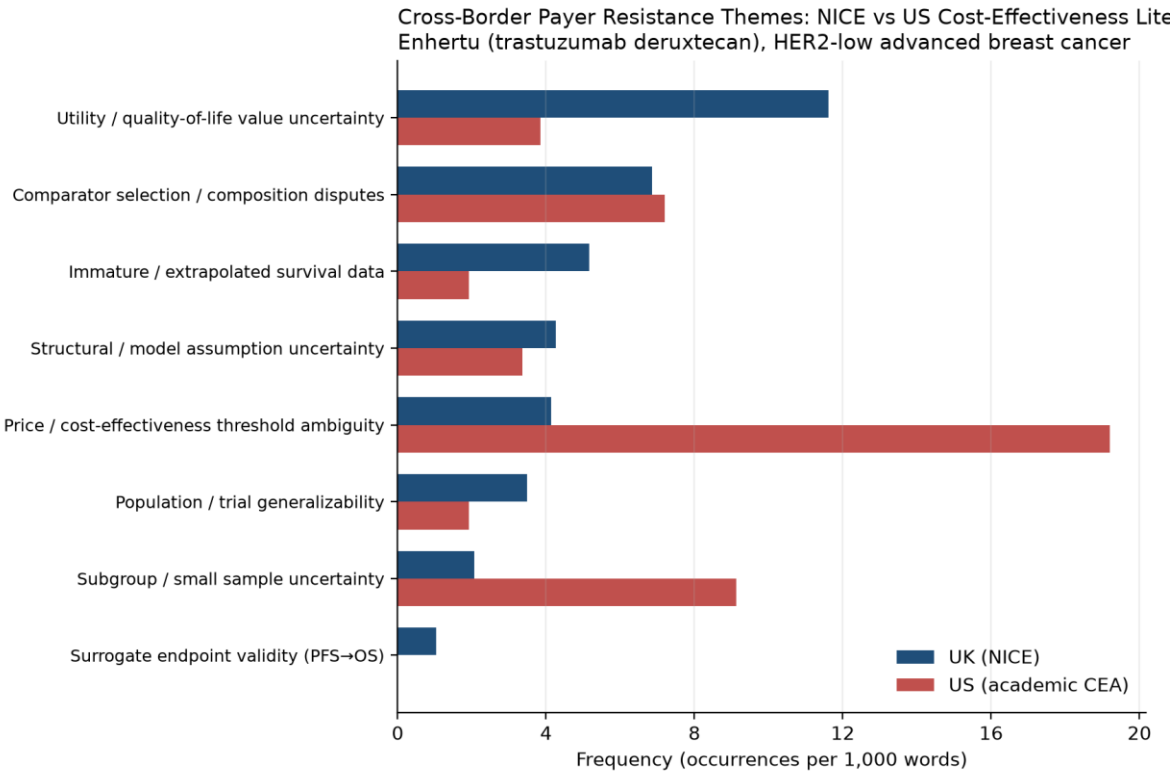


Figure 5: 3.2_chart_uk_vs_us_theme_comparison.png. Normalized theme frequency (per 1,000 words), NICE appraisals vs. US academic CEAs. See methodology and per-document limitations below.

Theme	UK (NICE)	US (academic CEA)	Direction
Price/cost-effectiveness threshold ambiguity	2.45	10.09	US>4x
Subgroup / small sample uncertainty	1.94	9.13	US>4x
Utility / QoL value uncertainty	7.62	2.88	UK>2x
Immature / extrapolated survival data	5.17	1.92	UK>2x
Comparator selection / composition disputes	6.85	7.20	~parity
Surrogate endpoint validity (PFS to OS)	0.77	0.00	UK only

Two sources of divergence can be identified. NICE appraisals focus on evidence quality and durability, including survival data immaturity, utility value uncertainty, and surrogate endpoint validity, all of which are UK weighted. US academic CEAs focus on statistical and pricing uncertainty, with threshold ambiguity and subgroup and small sample caveats dominating, likely reflecting the WTP threshold framing ($100,000–$200,000/QALY bands) that structures US papers but has no direct equivalent in NICE. Comparator selection is discussed at similar intensity in both bodies (6.85 vs. 7.20), suggesting a shared methodological concern rather than a jurisdiction specific one.

Limitations

- There are n=4 documents (2 UK, 2 US). This means that this is a descriptive comparison of specific texts, not a representative sample of NICE or US CEA literature generally. No claim is made about typicality beyond these four documents.
- Keyword-based extraction captures thematic mentions, not argumentative weight or outcome impact (a theme mentioned 18 times isn't necessarily 18x more decision-relevant than one mentioned once.)
- The two US papers themselves diverge sharply on some themes — most notably subgroup/small-sample uncertainty, where Peng et al. (2023) registers 16.87 mentions per 1,000 words against Shi et al.'s (2023) 0.99 (Figure Z, below). This means the "US" aggregate in the table above masks meaningful within-market variation on this theme specifically, and the aggregated figure shouldn't be read as implying a uniform US style.

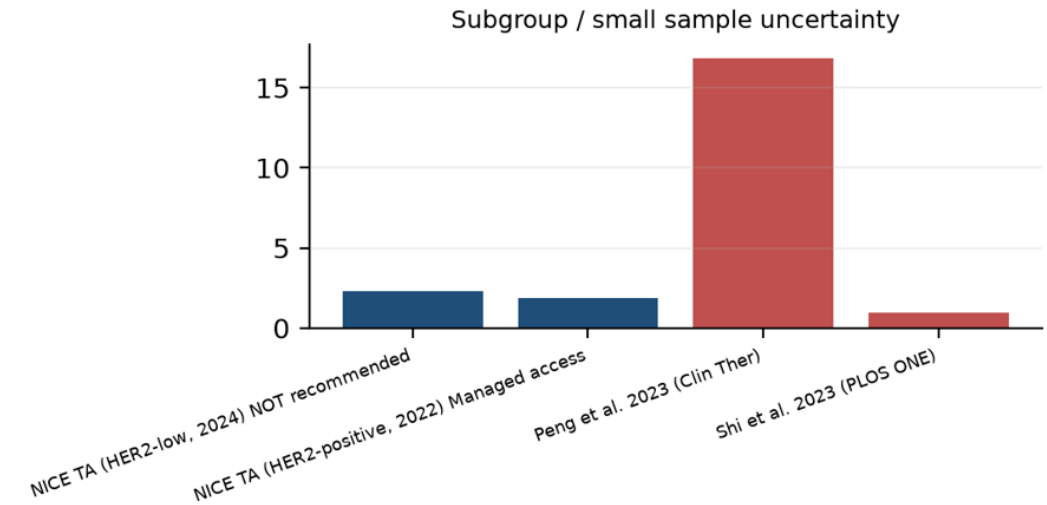


Figure 6: cropped subgroup-uncertainty panel from 3.2_chart_small_multiples_by_document.png. The 17x within-US gap between Peng and Shi shows the aggregate US figure (9.13/1,000 words) obscures substantial paper-to-paper variation.

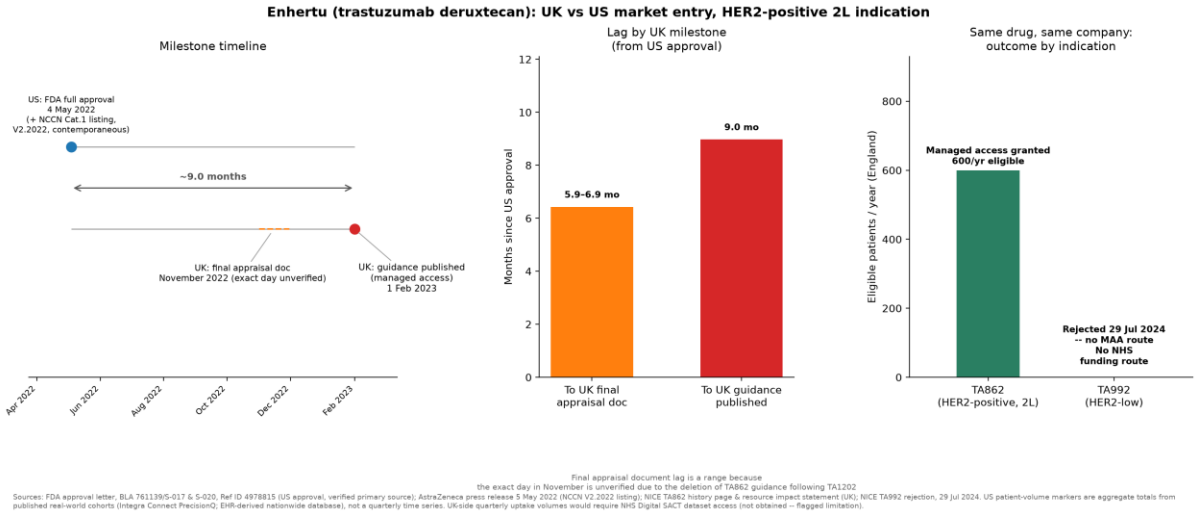
Summary of findings

The clearest, most defensible finding from this section is the surrogate-endpoint asymmetry, which is a binary presence/absence pattern (not a magnitude difference vulnerable to small-sample noise) and maps onto a real structural difference between NICE's HTA remit and US academic CEA conventions. The threshold-ambiguity and subgroup-uncertainty findings are directionally consistent with expectations about US payer-model literature, but they should be presented as observations from four specific documents rather than as a market-wide claim.

3.3: Market uptake

Headline finding

Trastuzumab deruxtecan (T-DXd) provides a focused case study of two distinct access frictions that apply even to a single medicine which has already been approved. The first is the interval between US regulatory approval and published NICE guidance, and the second is the degree to which reimbursement outcomes depend on indication rather than the medicine itself.



3.3_uptake_comparison.png. A three-panel comparison: (left) UK-vs-US milestone timeline, (middle) lag by UK milestone, (right) eligible-patient access by indication.

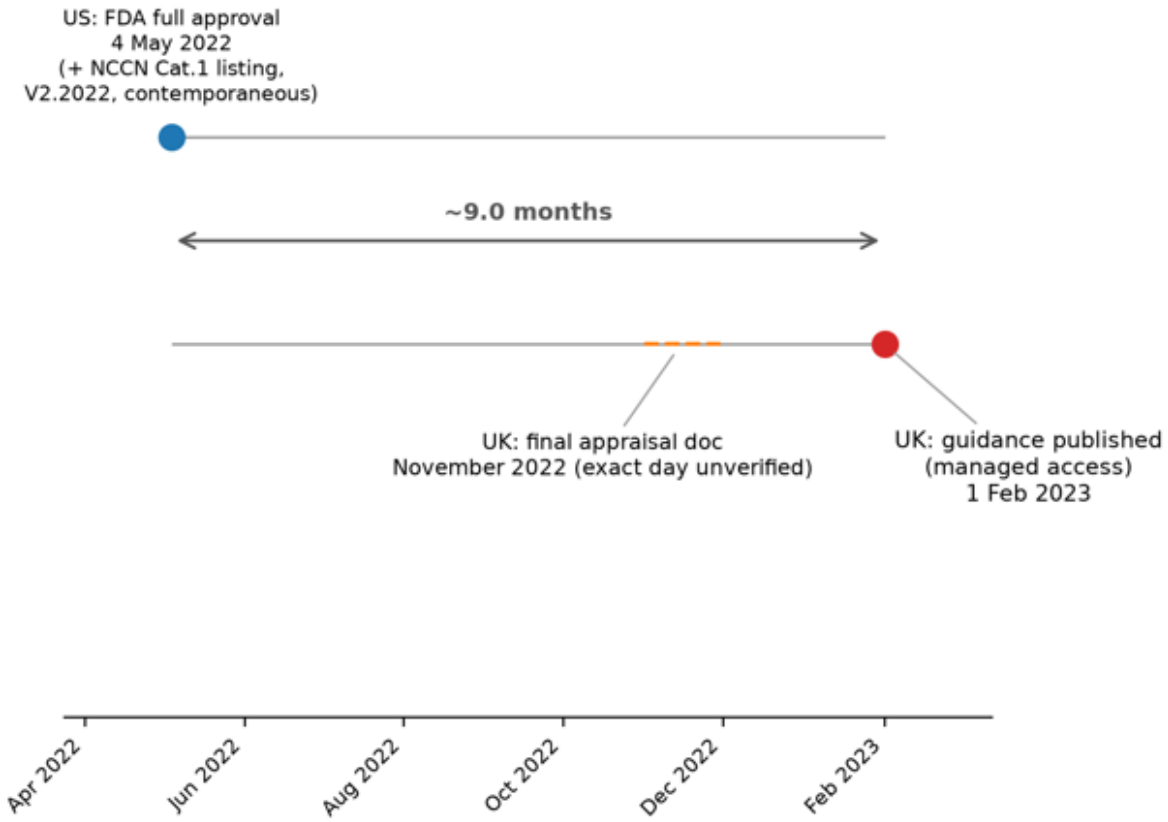


Figure 7a: US regulatory approval vs. UK reimbursement guidance timeline for T-DXd (HER2-positive, after 1 or more anti-HER2 treatments). FDA approval: 4 May 2022. NICE final draft guidance: November 2022.[footnote] NICE guidance published: 1 February 2023.

Panel 1: Milestone timeline

The FDA approved trastuzumab deruxtecan for HER2-positive breast cancer after prior anti-HER2 treatment on 4 May 2022. NICE issued final draft guidance in November 2022, and guidance was published on 1 February 2023, a gap of approximately nine months between US approval and published UK guidance.

Implication

FDA approval is necessary but not sufficient for NHS access. Formal NICE guidance was published nine months after US approval, so US patients had a licensed route to the therapy for a period in which no final NICE guidance existed in England. Publication of guidance is not the same as patient access, and interim Cancer Drugs Fund funding may have applied from the final draft stage, so the nine months should be read as a lag in formal guidance rather than a period with no access at all. For sponsors or investors modelling UK revenue timing against a US approval date, the interval is a risk to price in,

not a residual to adjust away. Because this is a single case, it should be tested against the wider appraisal dataset before being used as a planning benchmark.

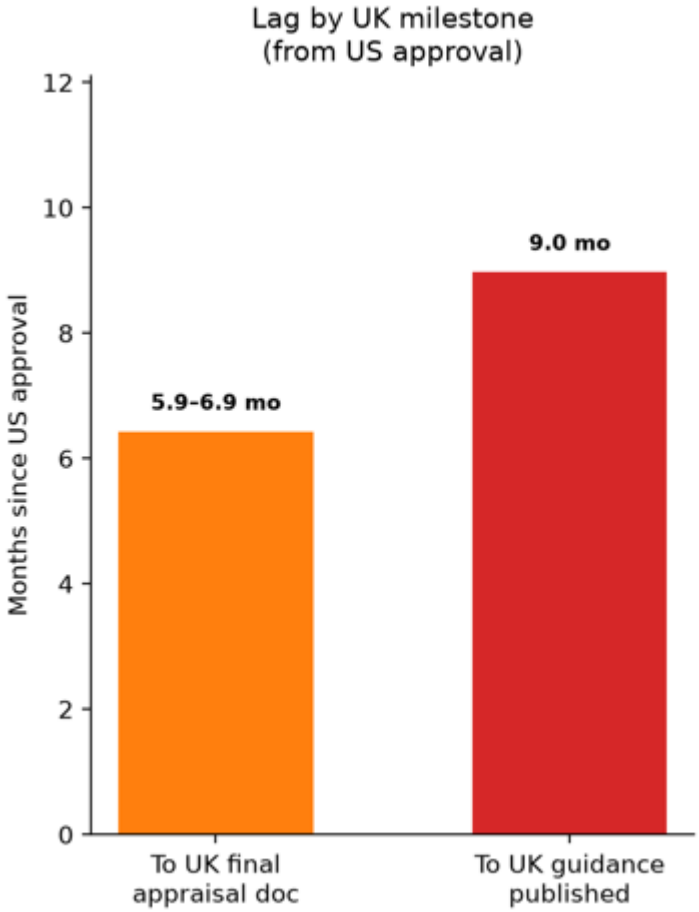


Figure 7b: Cumulative lag from US approval to UK appraisal milestones. About 6 to 7 months to NICE final draft guidance and 9.0 months to published guidance. Both are measured from US approval and not from the start of the NICE appraisal, so the pre-draft period is not all process time. Most of the interval elapsed before final draft guidance, not after it.

Panel 2: Lag by UK milestone

Measured from US approval, roughly 6 to 7 months passed before NICE issued final draft guidance and 9 months before guidance was published. The remaining 2 to 3 months between the two milestones is small relative to the period before final draft guidance. Both figures are measured from the FDA decision and not from the start of the NICE appraisal, which began earlier, so they should not be read as the duration of the appraisal itself.

Implication

This finding does not support a simple "publication lag" explanation, where delay is attributed to administrative processing after a decision has been reached. Most of the interval elapsed before final draft guidance, not after it. However, because the appraisal began before US approval, and UK marketing authorisation timing is not separated out here, the data cannot show how much of that period reflects appraisal process as opposed to regulatory sequencing. If the pattern holds across the wider dataset, it would suggest the access lag is structural rather than incidental, and that it may need addressing earlier in the pathway, such as at evidence readiness for submission, rather than through post-decision fixes.

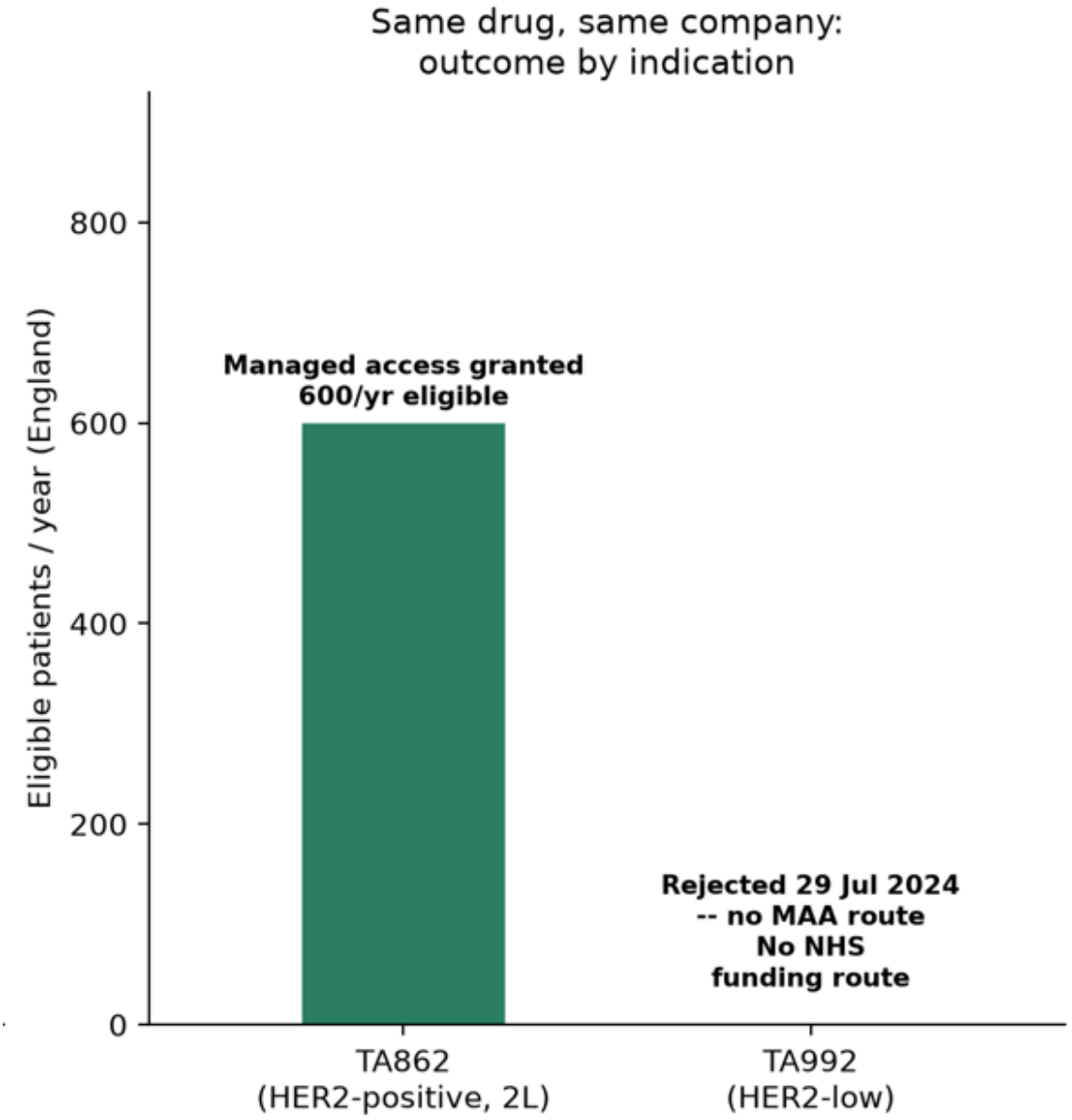


Figure 7c: NHS reimbursement outcome by indication for T-DXd. TA862 (HER2-positive, second-line): managed access, ~600 eligible patients annually. TA992 (HER2-low): rejected 29 July 2024, no managed access route.

Panel 3: Outcome by indication

TA862 (HER2-positive, second-line) secured managed access for approximately 600 eligible NHS patients annually. TA992 (HER2-low), an appraisal of the same medicine from the same manufacturer, was rejected on 29 July 2024, resulting in no managed access agreement and no NHS funding route. This leaves effectively zero eligible patients under that indication.

Implication

This is the most impactful finding of the section. The medicine, manufacturer and appraising body are held constant, yet the outcome differs by indication. The two appraisals also differ in trial evidence, comparator, population and date, so the contrast is consistent with indication-level cost-effectiveness driving the outcome but does not isolate it. It challenges any assumption that reimbursement status is portable across indications for a given therapy. A positive outcome in one indication carries little predictive value for a related but different indication, even where the underlying molecule is identical.

Key limitation

This single-medicine, two-indication case study is not representative of the broader dataset. The regression diagnostics in Section 3.1 do not apply to this analysis. Its value lies in illustrating access lag and divergent indication outcomes that may recur elsewhere in the dataset, not in establishing their frequency or magnitude.

Core findings

Trastuzumab deruxtecan exhibits two access frictions distinct from those captured by approval status alone. The first is an approximate nine-month lag between US approval and published NICE guidance, most of which elapsed before final draft guidance, not after it. The second is a reimbursement outcome that varies from managed access to no routine funding, depending on indication despite an identical medicine and manufacturer. Whether these frictions are general features of NICE's process or specific to this case is addressed in the key limitation above.

Overall implication

This case surfaces two access frictions worth testing against the wider appraisal dataset. First, published NICE guidance can lag US approval by a materially long interval. Second, a positive reimbursement outcome for one indication carries no guarantee for another indication of the same medicine. Until validated at scale, sponsors and investors should treat both as case-by-case risks to underwrite, not planning benchmarks to assume.

3.4: Economic Framing, Cost Effectiveness thresholds (UK vs US)

Headline finding

Even after NICE's 2026 threshold revision and full currency conversion, NICE's most generous historical threshold remains substantially below the floor of the US ICER range. This is not the result of exchange rates or reporting conventions. It reflects a structural difference in stated willingness-to-pay for a QALY between the two systems.

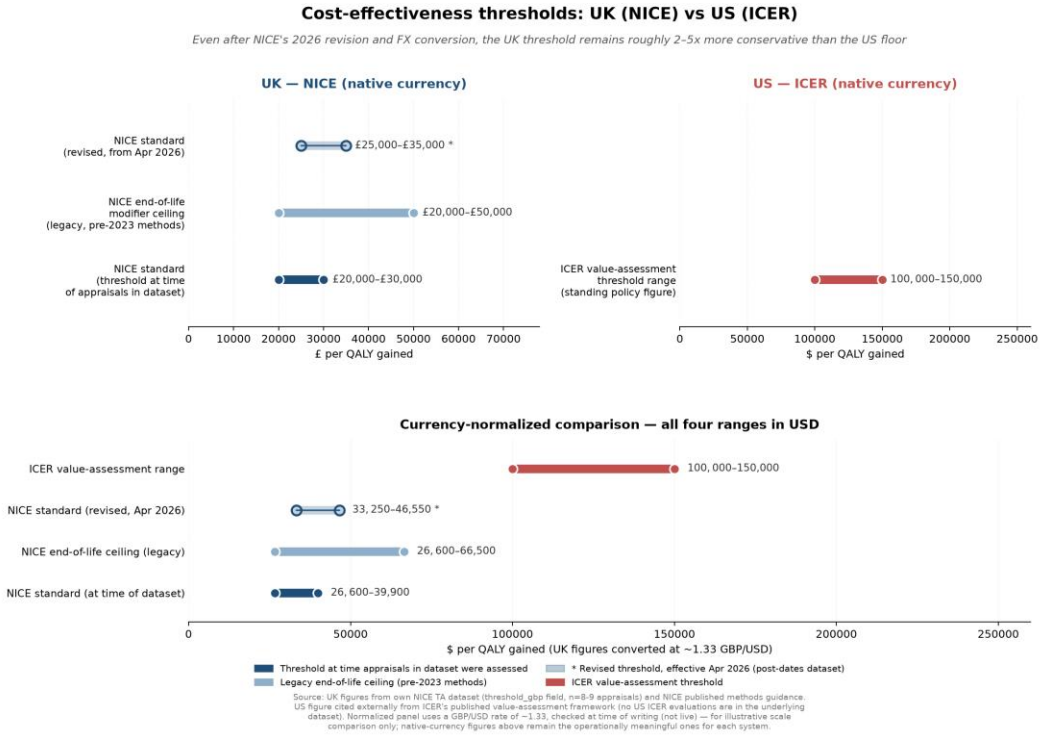


Figure 8: 3.4_threshold_comparison.png. A three-panel comparison: (top left) UK NICE thresholds in native currency, (top right) US ICER threshold in native currency, (bottom) all four ranges converted to USD for direct comparison.

Panel 1: UK (NICE) thresholds

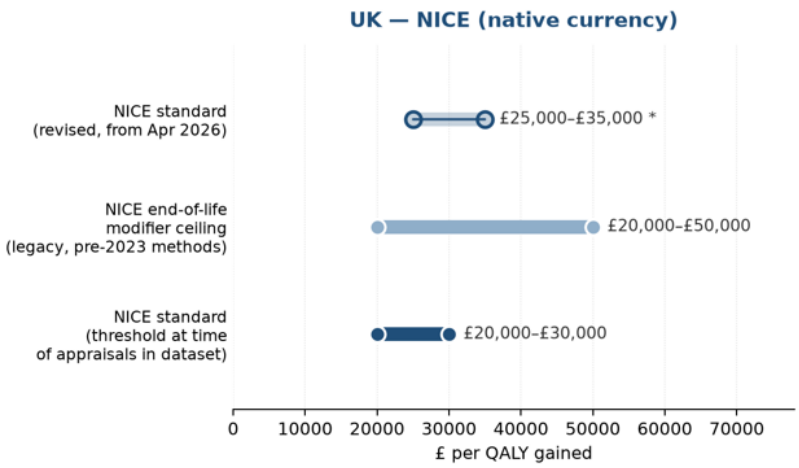


Figure 8a: UK (NICE) cost-effectiveness thresholds by policy regime, £ per QALY gained. Standard threshold (dataset period): £20,000–£30,000. Legacy end-of-life modifier: £20,000–£50,000. Revised standard (effective April 2026): £25,000–£35,000.

Three NICE thresholds are shown, all measured in £ per QALY gained. The standard threshold applied to most appraisals in this dataset was £20,000–£30,000, with treatments below £20k typically considered cost-effective and those above £30k increasingly difficult to justify. A wider legacy end-of-life modifier, £20,000–£50,000, applied under NICE's pre-2023 methods, allowing higher ICERs to be accepted where a treatment extended life near its end. Following NICE's 2026 methodology update, the standard threshold rises modestly to £25,000–£35,000.

Implication

The 2026 revision represents an incremental loosening, not a step-change. NICE remains a comparatively conservative payer even under its most current guidance. The legacy end-of-life modifier is the single widest UK range in absolute terms, but it applied only to a defined subset of appraisals meeting specific end-of-life criteria, not as a general standard.

Panel 2: US (ICER) threshold

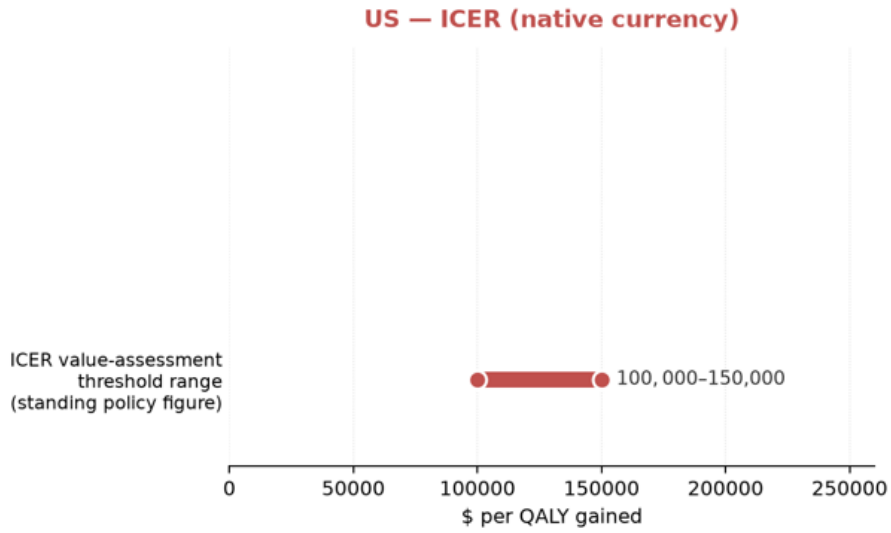


Figure 8b. US (ICER) value-assessment threshold, $ per QALY gained. $100,000 ($150,000, cited as a default benchmark rather than a binding national reimbursement threshold.)

The US panel shows a single range, $100,000 to $150,000 per QALY, the value-based price benchmark ICER has used in its own methodology since at least 2017, held constant through its 2019 and 2020 framework updates[26]. Unlike NICE's threshold, this is not a binding national reimbursement gate, as the US has no single mandated cost-effectiveness ceiling, but the ICER range functions as a standard benchmark widely referenced in US health economic evaluations, with independent analysis of the broader cost-effectiveness literature confirming this range has progressively displaced the historically dominant $50,000/QALY threshold as the standard US reference point since ICER's 2017 adoption[27].

Implication

This distinction matters for interpretation. A UK appraisal that breaches its threshold is formally rejected, whereas a US analysis that breaches ICER's benchmark carries no equivalent binding consequence. The two ranges sit at different points on a spectrum from advisory to enforceable.

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[26] Institute for Clinical and Economic Review (ICER) (2020) 2020-2023 Value Assessment Framework. Neumann, P. J., & Kim, D. D. (2023). Cost-effectiveness Thresholds Used by Study Authors, 1990-2021. *JAMA*, 329(15), 1312–1314. Citation of a $100,000/QALY threshold among US cost-effectiveness studies rose from 9.1% (1990–1999) to 47.0% (2020–2021), overtaking the $50,000/QALY benchmark that had dominated the 1990s–2000s, as $150,000/QALY citation grew in parallel from 0% to 23.6% over the same period.

Panel 3: Currency-normalized comparison

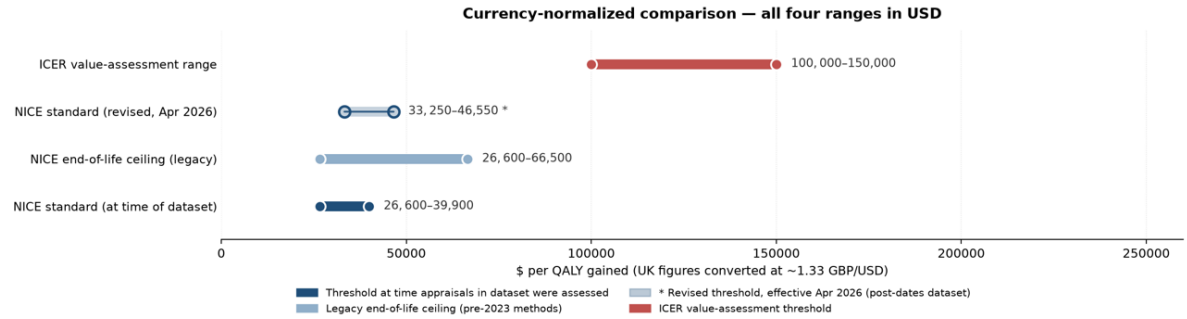


Figure 8c: Currency-normalized comparison of all four threshold ranges (USD, GBP/USD ≈ 1.33). Even the widest UK range (legacy end-of-life ceiling, $26,600–$66,500) does not reach the ICER floor ($100,000).

Converting all four ranges to USD at approximately £1 = $1.33 removes currency as a confounding factor. Results are as follows: NICE standard (dataset) becomes $26,600–$39,900. NICE revised (2026) becomes $33,250–$46,550. The NICE legacy end-of-life ceiling becomes $26,600–$66,500. The ICER range remains $100,000–$150,000.

Implication

Even the widest and most generous UK figure (the upper bound of the legacy end-of-life ceiling at $66,500) falls roughly a third short of the ICER floor. No UK threshold in this dataset, however broadly interpreted, approaches the US range at any point. The gap narrows slightly under the 2026 revision but remains wide: NICE's revised upper bound ($46,550) is still under half the ICER floor.

Key limitation

The GBP-USD exchange rate used (≈1.33) is a point-in-time snapshot, not a live or averaged rate, and was selected for illustrative scale comparison rather than financial precision. More importantly, the US and UK figures are not directly comparable policy instruments. NICE's threshold is a formal input into a binding reimbursement decision, while ICER's range is a widely cited but non-mandatory reference point. The comparison illustrates relative scale of willingness-to-pay, not two directly interchangeable decision rules.

Summary of findings

The clearest finding from this section is the persistent and substantial gap between UK and US cost-effectiveness thresholds, which holds even under the most favourable UK comparator (the legacy end-of-life ceiling) and after NICE's 2026 upward revision. This gap should be interpreted as evidence of differing stated willingness-to-pay between the two systems, with the limitation that the US figure is advisory rather than binding, and the UK figures span multiple distinct policy regimes (standard, legacy end-of-life, and post-2026) rather than a single constant threshold.

Section 4: Strategic Implications

This section translates the findings of Sections 3.1–3.4 into forward-looking implications for a company navigating a launch comparable to trastuzumab deruxtecan (T-DXd) a therapy with multiple indications, seeking reimbursement in both the UK and US within a similar timeframe. Each subsection restates the relevant finding briefly, states the mechanism connecting it to a strategic decision, and states the implication for a sponsor or investor. Consistent with Sections 3.1–3.4, single-case observations (drawn from the T-DXd case study in 3.3–3.4) are flagged as such and are not treated as generalizable on their own. Where a finding rests on the wider n=13 diagnostic set (3.1) or the four-document thematic comparison (3.2), that broader (but still limited) evidence base is also carried forward rather than upgraded.

4.1 EAG disagreement as an early diligence signal

Headline finding

Section 3.1 identifies EAG disagreement as the strongest observed predictor of NICE outcome across the verified n=13 dataset, with a monotonic relationship between disagreement level and outcome severity, though the small sample means this should be read as a hypothesis-generating pattern rather than a validated predictive relationship.

Mechanism

The EAG's assessment is produced before the final appraisal committee decision, and is often available, in preliminary form, well before NICE's guidance is published. If disagreement level tracks outcome as consistently as this dataset suggests, then a sponsor's own visibility into the EAG's reaction to its submitted model constitutes an early, internal signal of appraisal risk, arriving months ahead of the public decision.

Implication

For a company preparing a NICE submission, the EAG's response to the initial company model is a diligence checkpoint worth weighting heavily rather than a procedural step to clear on the way to committee. Where a comparable treatment shows significant early disagreement, particularly around the assumptions most likely to recur across NICE submissions (survival extrapolation, comparator selection), a sponsor has a window to revise its evidence package, adjust its commercial forecast, or open pricing discussions before the committee decision closes off those options. This does not predict outcome with confidence at n=13, but it identifies where diligence attention is best spent during the appraisal window itself. In practical terms, a preliminary EAG response coded as substantial disagreement should trigger an internal pricing and evidence review within the sponsor organisation immediately, rather than waiting for the final appraisal, since the EAG's reaction is available months before the committee's binding decision and this dataset's monotonic pattern suggests that window is where remaining leverage exists.

4.2 The surrogate endpoint blind spot in evidence package design

Finding

Section 3.2 finds that surrogate endpoint validity (progression-free survival as a proxy for overall survival) is a live, recurring point of contest in both NICE appraisals examined, while it is entirely absent from both US academic cost-effectiveness papers, a categorical absence rather than a graded difference across n=4 documents.

Mechanism

US academic cost-effectiveness models typically take a trial's published endpoint as an input rather than evaluating its validity as an outcome proxy. NICE's appraisal remit requires the committee to explicitly assess whether the surrogate reliably predicts the outcome that matters for reimbursement. A sponsor whose US-facing evidence package is built to satisfy academic CEA conventions may therefore never generate the specific PFS to OS extrapolation evidence a UK appraisal committee will independently demand.

Implication

Evidence packages built primarily against US conventions carry a structural gap when reused for a NICE submission, and this gap concerns exactly the kind of evidence that is costly to generate after trial design is fixed (long-term follow-up data supporting survival extrapolation). A sponsor targeting both markets should treat UK-style surrogate endpoint justification as a distinct deliverable to plan for at trial design stage, not a component that falls out of a US-oriented evidence package. In practical terms, OS follow-up collection should be written into the trial protocol at design stage, with a pre-specified interim OS analysis timed well ahead of trial closeout since Section 2.4's own case shows the interval

between primary (PFS-driven) completion and full trial closeout runs 4.5 to 5 years for T-DXd's own pivotal trials (DESTINY-Breast03 2021-05-21 to 2026-07-30 DESTINY-Breast04 2022-01-11 to 2026-08-01), and NICE's TA862 guidance was published just under 21 months after DESTINY-Breast03's primary completion, using OS data that was, by construction, still immature.

This finding is drawn from four documents and should be interpreted as illustrative of a real structural difference between the two evidence regimes, not as a claim about the average US CEA or the average NICE appraisal.

4.3: Indication-level transfer risk

Finding

Section 3.3 shows that T-DXd received managed access for its HER2-positive indication (TA862) while its HER2-low indication (TA992) was rejected outright, with the medicine, manufacturer, and appraising body held constant across both. Indication was the only variable that differed.

Mechanism

NICE evaluates cost-effectiveness at the level of the indication-specific evidence package (population, comparator, trial data, and modelled cost-effectiveness for that specific use), not at the level of the molecule. Therefore, a reimbursement decision for one indication provides no structural guarantee for another, even where a prior positive decision might create an expectation of consistency.

Implication

A sponsor or investor with a multi-indication launch plan should underwrite each indication's reimbursement outcome independently rather than treating an earlier positive decision as a track record that de-risks subsequent submissions for the same molecule. This has direct consequences for how a company sequences indication launches and how an investor should discount projected revenue from indications that have not yet been appraised, even where the lead indication has already cleared. In practical terms, revenue models should assign each unappraised indication its own independent reimbursement probability rather than inheriting the lead indication's outcome as a prior, and the T-DXd case sets the scale of what's at stake if that discipline is skipped a swing from roughly 600 eligible NHS patients annually to zero, on the same molecule, from the same manufacturer, decided on indication alone. As with 3.3's own conclusion, this is a single-medicine, two-indication case study, and its value is in demonstrating that this risk exists and can be severe

(a shift from ~600 eligible patients annually to zero), not in establishing how frequently it recurs across the broader appraisal landscape.

4.4 Threshold gap and launch sequencing

Finding

Section 3.4 shows that even NICE's most generous historical threshold, converted to USD, falls short of the floor of the US ICER benchmark range, a gap that narrows only modestly under NICE's 2026 revision.

Mechanism

NICE's threshold functions as a binding constraint on a formal reimbursement decision, while ICER's range is an advisory reference point with no equivalent enforcement mechanism in the fragmented US payer system. A price that would comfortably clear an implicit US benchmark can still fail a UK appraisal on cost-effectiveness grounds alone, independent of clinical merit.

Implication

A single global list price calibrated to US willingness-to-pay carries meaningful UK rejection risk, and the direction of this risk is asymmetric. A price low enough to clear NICE's threshold is very unlikely to be a binding constraint in the US, whereas the reverse is not true. This favours launch sequencing and pricing strategies that treat the UK threshold as a design constraint from the beginning, for example through UK-specific pricing agreements or managed access arrangements, rather than as a market to be revisited only after a US-anchored price has already been set. In practical terms, global list price should be stress-tested against NICE's post-2026 ceiling (~$46,550) before it is tested against the ICER benchmark floor ($100,000), because a price that clears NICE will almost always clear the advisory US range by a wide margin, while the reverse routinely fails so the UK figure, not the US figure, should anchor the pricing exercise, with UK-specific agreements (e.g. managed access, confidential discounts) built in from the start rather than negotiated reactively after a NICE rejection. As Section 3.4 notes, ICER's range is advisory and the two figures are not directly interchangeable decision rules, so this implication concerns relative scale of willingness-to-pay rather than a precise transferable price ceiling.

4.5 Synthesis

Interpreted together, Sections 3.1–3.4 point to a consistent theme rather than four unrelated findings. UK and US market access show different dynamics, not merely different timelines or different numbers. The EAG disagreement signal (4.1) and the surrogate

endpoint gap (4.2) both concern how UK evidence standards diverge from US-oriented evidence generation, upstream of any pricing decision. The indication transferability finding (4.3) and the threshold gap (4.4) both concern how a positive US-facing outcome, whether regulatory approval or an implicit pricing benchmark, fails to transfer cleanly to the UK, after the initial launch decision.

For a company planning a comparable launch, the combined implication is that UK market access risk cannot be effectively managed as a later-stage adaptation of a US-anchored strategy. The points examined here (evidence package design, EAG engagement, indication sequencing, and pricing architecture) each show a UK-specific consideration that is cheapest to address early and most expensive to address after key decisions, such as trial design or headline pricing, have already been made for a US-facing strategy.

Key limitation

This section synthesizes findings from a diagnostic dataset of n=13 appraisals (3.1), a four-document thematic comparison (3.2), and a single-medicine, two-indication case study (3.3–3.4). None of these individually supports a general claim about NICE or US payer behaviour at a market-wide level, and Section 4 does not treat them as such collectively. The value of this section is in showing that these four findings, each independently limited, point in a consistent strategic direction. It is not a claim that this direction has been established with statistical confidence, and it should be read as a set of risk factors to underwrite and test further rather than a validated playbook.

Summary of findings

The clearest strategic conclusion from this section is that UK and US market access diverge structurally rather than incidentally, across evidence standards (4.2), decision-making signals (4.1), portability of outcomes (4.3), and pricing logic (4.4). A sponsor or investor evaluating a comparable launch should treat UK-specific risk as a first-order planning input from trial design onward, not a market-access workstream to be resolved after a US-focused strategy is already set.

Methodology appendix

Summary

This project draws on three distinct evidence bases, each with a different sample size and a correspondingly different evidentiary weight: a verified n=13 dataset of NICE technology appraisals for HER2-positive and HER2-low breast cancer therapies (Section 3.1). The second is a four-document thematic comparison of two NICE appraisals against two US academic cost-effectiveness papers (Section 3.2). The third is a single-medicine, two-indication case

study of trastuzumab deruxtecan across UK and US markets (Sections 3.3, 3.4, and 2). Two methodological decisions shape how these evidence bases were used and are worth noting before the detailed sections below. First, Section 3.1 was deliberately built as a descriptive diagnostic rather than a predictive model, because the sample size could not support the latter without producing a misleading appearance of precision. Second, the project's illustrative case study anchor shifted from TA992 (HER2-low, rejected) to TA862 (HER2-positive, managed access) partway through, for reasons of evidential completeness rather than convenience. Both decisions are documented in full below (Sections B and C). Readers satisfied with this summary can treat the findings in Sections 3.1–3.4 and 4 as building on a verified but small-sample evidence base, with generalizability explicitly bounded at each section. Readers who want the underlying reasoning should continue to the relevant subsection.

A. Dataset construction and verification

The core dataset (n=13) comprises HER2-positive and HER2-low breast cancer therapy appraisals published by NICE, extracted with the following fields: appraisal ID, indication, EAG disagreement level (aligned / partial / substantial, coded ordinally as 0/1/2), end-of-life modifier status, uplift-applied status, era (pre- or post-2022 EoL methodology), base-case ICER (where disclosed), and final committee outcome. Each record was independently verified against NICE's published guidance documents rather than taken from a secondary aggregator, using purpose-built verification scripts (“verify_trial_data.py”, “verify_ta992_timeline.py”, “verify_cms_j9358.py”) that cross-checked extracted values against primary source text. This verification step surfaced and corrected several extraction errors during the project's early stages, which is the reason a dedicated verification pass was retained as a standing step rather than a one-off check.

Two fields warrant a specific note on completeness. The "uplift_applied" field is coded from the appraisal outcome itself (an uplift is recorded as insufficient only where the outcome was already negative), which makes it circular by construction. It is retained in the dataset for transparency but is excluded from any claim about predictive relationships, as stated in Section 3.1. Base-case ICER is usable as a point estimate in only 5 of 13 appraisals. The remainder are confidential, undisclosed, reported only as a lower bound, or multi-valued. This reflects NICE's public disclosure practice rather than a gap in extraction and constrains any future use of ICER magnitude as a covariate.

B. Section 3.1: the predictive-to-descriptive pivot

Section 3.1 was originally scoped as a predictive (Gradient Boosting or Random Forest) model of NICE appraisal outcome. This approach was ultimately not used, and the reasoning is significant for how Section 3.1's findings should be interpreted.

A logistic regression on the available covariates (n=11, excluding the two CDF-managed-access rows) achieved 100% in-sample classification accuracy, with coefficients diverging to approximately −16 (EAG disagreement) and +10 (end-of-life status). This complete separation is a known attribute of maximum-likelihood estimation at small sample sizes. The model has enough flexibility to perfectly separate outcome classes by chance, with no held-out data to test whether that boundary generalises.

Two checks confirmed this. Leave-one-out refitting across all 11 observations showed large swings in coefficient magnitude (EAG disagreement from −8.4 to −16.3, end-of-life from +7.8 to +11.2) while sign remained stable. The direction stayed consistent, the magnitude was unstable, and therefore predictions were unstable for any new case. An events-per-variable calculation against standard guidance (10 to 20 minority-class events per predictor) showed the dataset's 4 rejected appraisals fall 6 to 16 cases short of the minimum needed for a 2-predictor model, implying a dataset 5 to 15 times the current size would be required before a predictive model of this type could be responsibly fit.

Given this, the regression was retained as a diagnostic exercise rather than reported as a substantive finding, and Section 3.1 was reframed around the descriptive pattern of outcome by EAG disagreement level instead. Due to this, Section 3.1 presents a monotonic association with an explicit hypothesis-generating limitation, rather than predicted probabilities or model-derived risk scores, and why it is more cautious than a typical predictive-modelling section.

C. Shift in Anchor Case from TA992 to TA862

The illustrative case study in Sections 3.3–3.4 was originally anchored on TA992 (HER2-low), as the more recent decision at the time. TA992 was rejected outright, with no managed access route agreed. This is a valid data point, but one that could not support Section 3.3's access-lag and administrative-timeline analysis, since a rejected appraisal has no guidance publication date to measure lag against, and a single rejected case cannot illustrate the indication-transfer contrast central to that section.

The anchor was therefore shifted to TA862 (HER2-positive, second-line), which reached managed access and therefore, carries a full set of UK timeline milestones to compare against the US approval date. TA992 was retained, but repositioned as the contrast case in Panel 3 (outcome by indication), which allows Section 3.3 to isolate indication as the variable that caused the reimbursement outcome to diverge for an otherwise identical medicine, manufacturer, and appraising body. The same shift carries into Section 3.4, where TA862's cleared threshold (not TA992's rejection) anchors the UK-side cost-effectiveness comparison against the US ICER benchmark.

This is noted explicitly because it determines how Sections 3.3–3.4 should be interpreted. TA862 is the headline case in both, with TA992 functioning as the contrast case motivating Section 4.3's indication-transferability finding, not as an independent result.

D: Theme extraction methodology (Section 3.2)

Eight resistance and uncertainty themes (comparator selection, survival data immaturity, population generalizability, cost-effectiveness threshold ambiguity, structural and model assumptions, subgroup uncertainty, surrogate endpoint validity, and utility/QoL uncertainty) were defined prior to extraction, based on a review of common contestation points in HTA and cost-effectiveness literature, and then applied via keyword matching (theme_extraction.py) across four source documents, namely two NICE technology appraisals for T-DXd (HER2-low, not recommended, 2024; HER2-positive, managed access, 2022) and two US academic cost-effectiveness papers (Peng et al., 2023; Shi et al., 2023). Raw counts were normalised per 1,000 words to control for length differences between UK appraisal documents (3,955 and 3,787 words) and US academic papers (1,067 and 1,015 words). Without this normalisation, the longer UK documents would mechanically register higher raw theme counts regardless of true thematic emphasis.

An initial counting error, in which certain theme-keyword matches were double-counted across overlapping keyword sets, was identified during quality review and corrected before the final figures and write-up were produced. This is noted for transparency. The counts and charts in Section 3.2 reflect the corrected extraction.

Keyword-based extraction captures thematic mentions, not argumentative weight, and a theme mentioned multiple times is not necessarily proportionally more decision-relevant than one mentioned once. This is stated as a limitation in Section 3.2 and is repeated here because it bears on how the aggregated UK-vs-US comparison (Figure 5) should be interpreted. The four-document aggregate is a description of these specific texts, not a representative sample of NICE or US CEA literature in general, and the paper-to-paper variation documented within the US pair (Figure 6) illustrates why the aggregate figure alone would overstate uniformity if interpreted without that limitation.